

Arpan Pathak

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SUMMARY

Senior Software Engineer, **8 years**: developer tools, observability platforms, and performance-critical distributed systems at **Amazon, Microsoft, Oracle**. Full-stack **Rust, C++, Kotlin, Go, Python, JavaScript**; **Kubernetes, eBPF/Cilium**, GPU/CUDA, cloud **performance optimization**.

EXPERIENCE

Software Developer II Microsoft, Redmond, WA Jun 2025 – Aug 2026

- **Cluster Observability: Prometheus + Azure Monitor**, blue-green deployments, automated validation across **AKS** clusters in France and Germany.
- **Kubernetes Operators & eBPF/Cilium**: Authored **Kubebuilder operators** reconciling network-policy state at scale; eBPF/Cilium-based L7 filtering and network visibility.
- **C++ → Rust Migration**: Rebuilt the Information Protection service from **legacy C++/C#** in **Rust**, moving memory safety to compile time; **~40% lower P99**, **2x throughput**, **3x lower memory/request**.
- Stack: **Rust, C++, eBPF, Cilium, Kubernetes, Linux**

Senior Software Engineer Oracle Cloud Infrastructure, Seattle, WA Oct 2024 – Mar 2025

- **Developer-Facing SDK (Go)**: Built the **Terraform provider** for Oracle Autonomous Database, replacing legacy control-plane tooling; **\$1.2M/month projected savings**.
- **Database Engine Internals**: Optimized **lock-free data structures** in a proprietary DB engine, achieving **+45,000 ops/s** in high-concurrency read-write paths.
- Stack: **Java, Go, Terraform, OCI**

Software Development Engineer II Amazon, Seattle, WA & Hyderabad, India Mar 2021 – Oct 2024

- **Real-Time ML Ranking (Python/PyTorch)**: **XGBoost + deep-learning ensemble** at **100ms** across FreeVee, Twitch, Prime Video; **5% CTR lift**, **\$5.4M/month revenue growth**.
- **BERT NLP Service**: **5,000 QPS** at peak; cut support-ticket triage SLA from **7 days to 2 hours**.
- **Full-Stack Developer Platforms**: **React.js microfrontend** migration (deploy **5 days** → **30 min**); event-bus **SDK** (Kotlin, **<50ms**) used by publisher teams.
- **Data Platform**: Built a **1-petabyte** warehouse/lake with **Spark (AWS Glue)** pipelines for financial reporting and BI.
- Stack: **Kotlin, Java, Python, PyTorch, Spark, Kafka, DynamoDB, Redshift, AWS**

SDE Razorpay, Bengaluru, India Aug 2020 – Feb 2021

- Go UPI Payment Gateway Aggregator: **1M+ txns/hr**, **99.99% uptime**, **150ms P99** at **50K TPM**.

Software Engineer Mindfire Solutions, Bhubaneswar, India Aug 2018 – Feb 2020

- Java/Spring full-stack tools, AR product-search microservice (**10K+ daily queries**); latency cut **5 min** → **1 min**.

PERSONAL PROJECTS

Driving-CivicSense: Edge AI Vision System (Rust)

- Real-time on-device perception pipeline in **Rust** on the **NVIDIA Jetson Orin Nano Super** (67 INT8 TOPS, 8 GB, 7–15 W): **YOLOv8/v11 INT8 ONNX (~12 ms)** + Deep SORT/Kalman tracking + formally-proven kinematic decision engine; **100%** on-device, privacy-first.
- Research paper: **Deterministic Intersection Blockage Prediction** | arpanpathak.github.io/driving-civicsense-vision-model/ | Code: github.com/arpanpathak/driving-civicsense-vision-model
- Books: **Seeing Machines** | arpanpathak.github.io/seeing-machines-book · **CUDA Kernels** (CUDA-Oxide / Rust-to-CUDA) | arpanpathak.github.io/gpu-parallel-book

SKILLS

Languages: Rust, C++, Go, C, Kotlin, Java, Python, TypeScript, JavaScript, Shell Script

Systems Programming & Performance: Lock-Free Data Structures, Concurrency, Memory Safety, Memory Profiling, Cache Optimization, Performance Engineering, Low-Latency Systems, OS Internals, Computer Architecture

Networking & Protocols: TCP/IP, UDP, TLS, mTLS, DNS, HTTP/2, HTTP/3, QUIC, WebSockets, RDMA, InfiniBand, DPDK, Socket Programming, Packet Processing, Network Performance Optimization

Network & Cloud Security: eBPF, XDP, Cilium, Kernel-Level Networking, VPC, Subnets, Routing, NAT, VPN, Peering, Load Balancing, Security Groups / Firewalls

Distributed Systems & Streaming: Consensus, Replication, Sharding, Apache Kafka, Apache Flink, Apache Spark, Spark Streaming, AWS Kinesis, Change Data Capture (CDC), Microservices

Cloud & Infra: AWS (EC2, ECS, Lambda, Kinesis, EMR, SageMaker), Azure (AKS), Google Cloud, Kubernetes (EKS, AKS), Docker, Terraform, Kubebuilder, Istio

Observability & Profiling: Prometheus, Grafana, Azure Monitor, AWS CloudWatch, Trace & Sampling Data, Performance Profiling

GPU & Parallel Computing: CUDA, CUDA-Oxide (Rust-to-CUDA), PTX, SIMT, GPU Kernel Programming, GPU-Accelerated Computing

ML Inference & AI: PyTorch, TensorFlow, XGBoost, Transformer Models (BERT), LLMs, ONNX Runtime, Edge/On-Device Inference, Model Pruning & Quantization, Computer Vision (YOLOv8/v11), CNN, NLP, Recommendation Systems

Data & Storage: DynamoDB, PostgreSQL, Redis, MongoDB, MySQL, Amazon Redshift, Amazon S3

APIs & Web: gRPC, REST API Design, Protobuf, React.js

Core CS: Data Structures & Algorithms, System Design, Operating Systems, Linux

EDUCATION

B.Tech, Computer Science & Engineering, RCC Institute of Information Technology, Kolkata | 2018